

Distance Weighting Algorithm for TuneTrace

April 3, 2025

1 Introduction

TuneTrace requires an effective distance weighting algorithm to compare melodies based on their direction, note values, and timing. To ensure a fair comparison, the Dynamic Time Warping (DTW) distances are first normalized by dividing them by the length of the user’s melody before applying the weighting function.

1.1 Definition of Direction

The *direction* of a melody describes the relative movement of successive notes. Given a sequence of notes $N_1, N_2, \dots, N_m$, the direction sequence d is defined as:

$$d_i = \begin{cases} 1, & \text{if } N_{i+1} > N_i \quad (\text{ascending}) \\ 0, & \text{if } N_{i+1} = N_i \quad (\text{same note}) \\ -1, & \text{if } N_{i+1} < N_i \quad (\text{descending}) \end{cases} \quad (1)$$

where d_i represents the movement between the i th and $(i + 1)$ th note. By comparing these directional sequences, we can assess how similar two melodies are in terms of contour.

1.2 Definition of Note Values

The *note value* represents the pitch of a note, mapped from classical notation (e.g., C2, G#1, Db0) to an integer using the MIDI numbering system. This transformation follows the formula:

$$\text{MIDI note} = \text{index}(\text{pitch class}) + (\text{octave} + 1) \times 12 \quad (2)$$

where the pitch class is indexed from the sequence:

$$[\text{C}, \text{C}\#, \text{D}, \text{D}\#, \text{E}, \text{F}, \text{F}\#, \text{G}, \text{G}\#, \text{A}, \text{A}\#, \text{B}] \quad (3)$$

1.3 Definition of Timings

The *timing* of a note represents the duration in seconds for which a note (or a rest) is held.

2 Distance Normalization and Weighting Function

The DTW distances for direction (d), note values (n), and timings (t) are first normalized by dividing by the length of the user's melody (L):

$$d' = \frac{d}{L}, \quad n' = \frac{n}{L}, \quad t' = \frac{t}{L} \quad (4)$$

Next, we apply the weighting function to each of these normalized distances:

$$D = \text{weighting_function}(d'), \quad N = \text{weighting_function}(n'), \quad T = \text{weighting_function}(t') \quad (5)$$

2.1 Constraints

The function $\text{weighting_function}(x)$ satisfies the following properties:

1. $\text{weighting_function}(0) = 0$
2. $\text{weighting_function}(x_0) = y_0$
3. For $x > x_0$, $\text{weighting_function}(x + n) \approx \text{weighting_function}(x) \cdot b^{(n/s)}$ for any $n \geq 0$
4. For $x < x_0$, $\text{weighting_function}(x) \approx 0$

2.2 Factorization

To meet these constraints, $\text{weighting_function}(x)$ is expressed as the product of multiple factors:

$$\begin{aligned} \text{Main Factor: } Mf(x) &= b^{(x-x_0)/s} \quad (\text{ensures exponential growth}) \\ \text{First Correcting Factor: } Fcf(x) &= \frac{1}{1 + a^{(x_0-x)/s}} \quad (\text{limits growth for small } x) \\ \text{Second Correcting Factor: } Scf(x) &= \frac{2}{1 + e^{-2x}} - 1 \quad (\text{ensures } \text{weighting_function}(0) = 0) \\ \text{Third Correcting Factor: } Tcf(x) &= \frac{2y_0(1 + e^{-2x})}{1 - e^{-2x}} \quad (\text{ensures } \text{weighting_function}(x_0) = y_0) \end{aligned}$$

Thus, the final function is given by:

$$\text{weighting_function}(x) = Mf(x) \cdot Fcf(x) \cdot Scf(x) \cdot Tcf(x) \quad (6)$$

2.3 Calculating Weighted Distances

By substituting d' , n' , and t' into the weighting function, we obtain the final weighted distances:

$$D = \text{weighting_function}(d'), \quad N = \text{weighting_function}(n'), \quad T = \text{weighting_function}(t') \quad (7)$$

These values are then used in the next step to compute the overall aggregated distance.

3 Combining Weighted Distances

The function `aggregate_weighted_distance( $D, N, T$ )` aggregates the separate weighted distances into a single value.

3.1 Definition

Given weight parameters W_d , W_n , and W_t , the final distance is calculated as:

$$\text{aggregate_weighted_distance}(D, N, T) = \frac{DW_d + NW_n + TW_t}{W_d + W_n + W_t} \quad (8)$$

4 Similarity Function

The similarity between two sequences is defined as:

$$\text{sim}(\text{aggregate_weighted_distance}(D, N, T)) = \frac{100\%}{1 + \text{aggregate_weighted_distance}(D, N, T)} \quad (9)$$

This ensures that as the weighted distance increases, the similarity percentage decreases smoothly.

5 Conclusion

This approach to distance weighting ensures a mathematically consistent and interpretable similarity measure, which is essential for accurate melody recognition. Future improvements may involve adaptive weighting schemes based on empirical testing.